

Properties of pure substance

Pure substance → A substance that has a fixed chemical composition throughout.

- It has homogeneous composition
- may exist in 2 or more phase but composition is same in all phases.

A mixture of liq. and water vapour is a pure substance but

A mixture of liq. and gaseous air is not.

Eg. → water (s, l & v phase)
CO₂
N₂

Sp. Volume → Vol. occupied by a unit mass of substance
 $\boxed{v}$

Sensible heating / specific heating

↓
heating that is used to increase the temp. of the substance without changing its phase

Eg. ↓

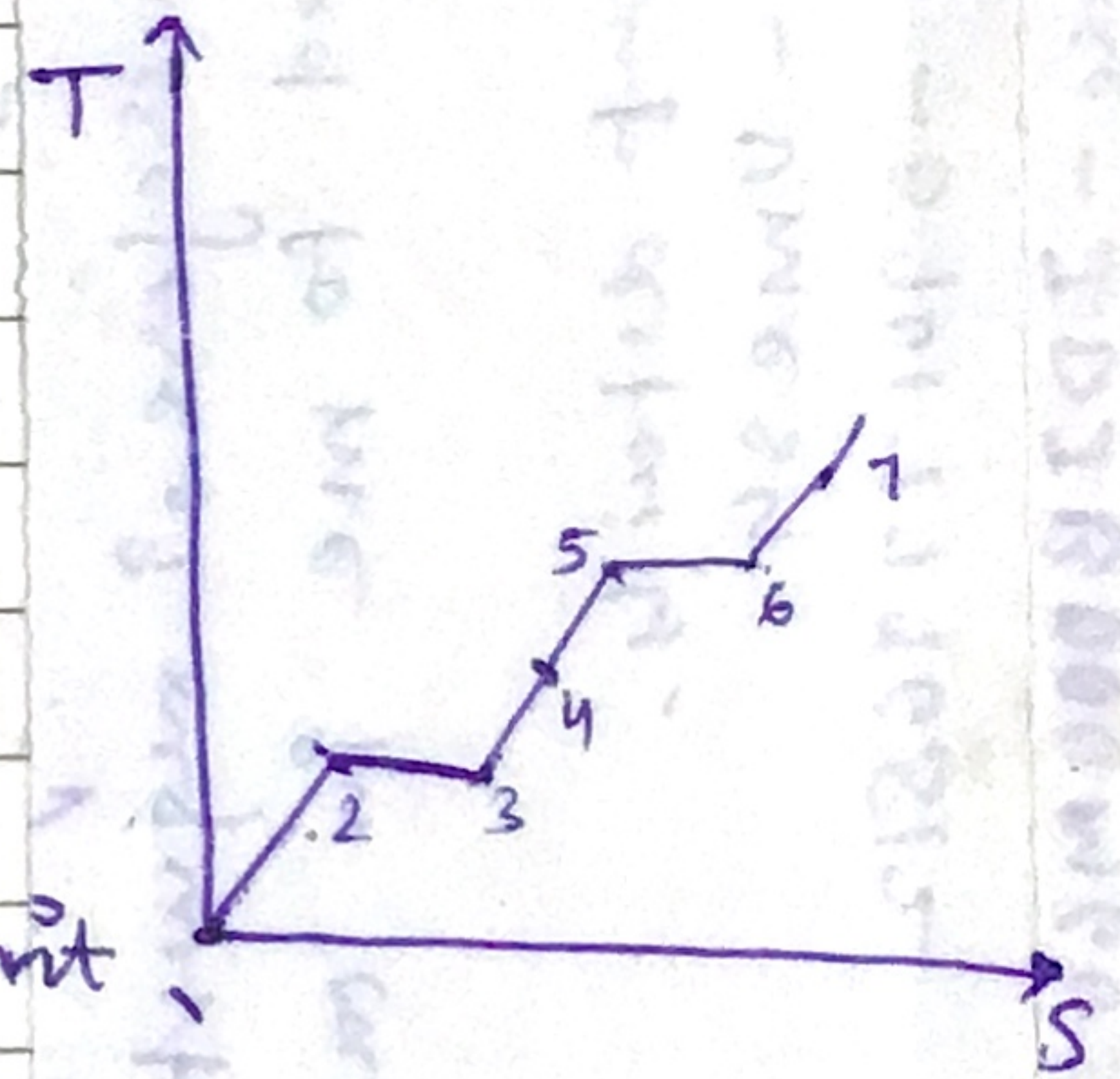
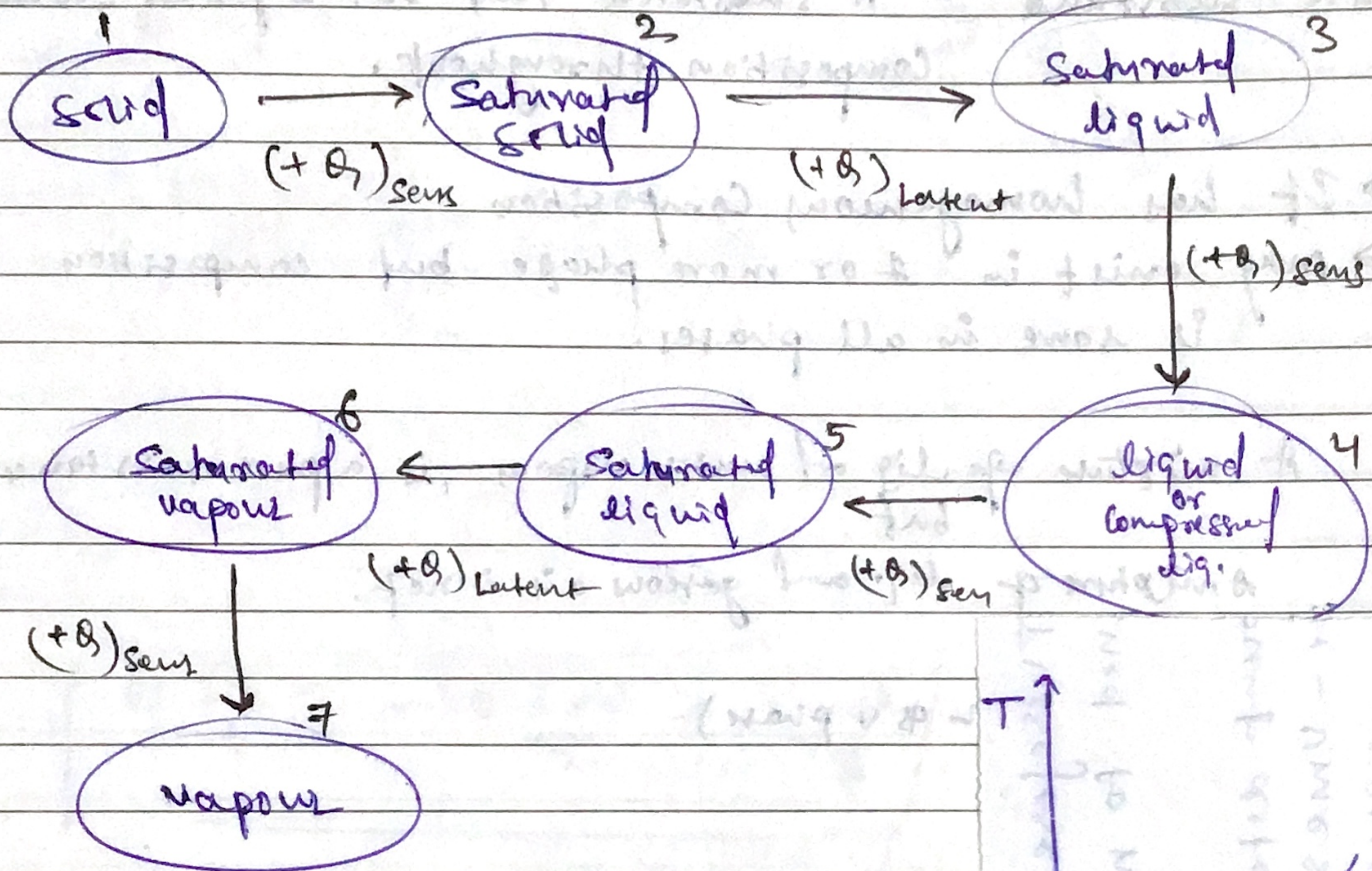
$$\boxed{Q = m C \Delta T}$$

m ← Mass of subst. (kg)

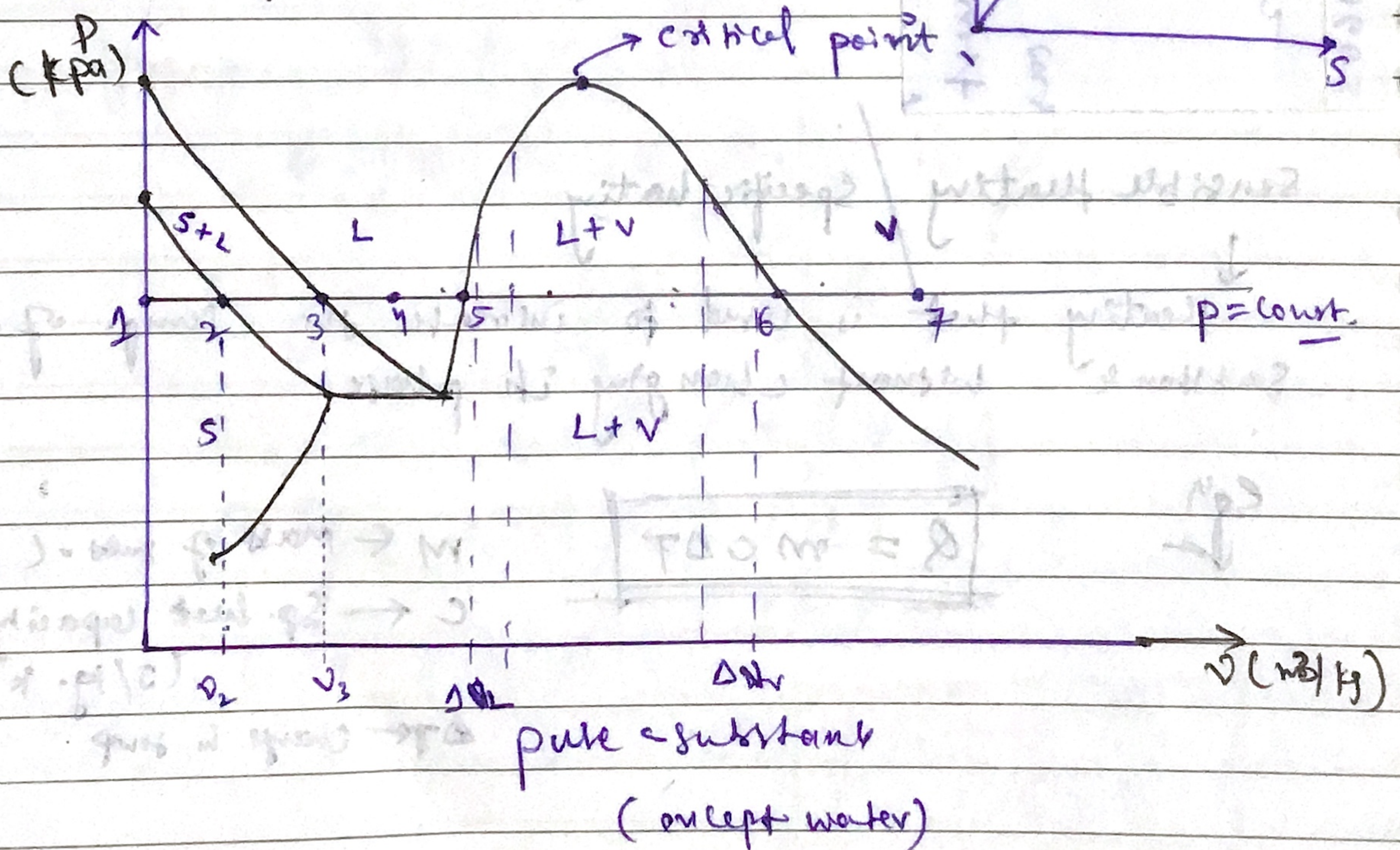
C ← Sp. heat capacity (J/kg·K)

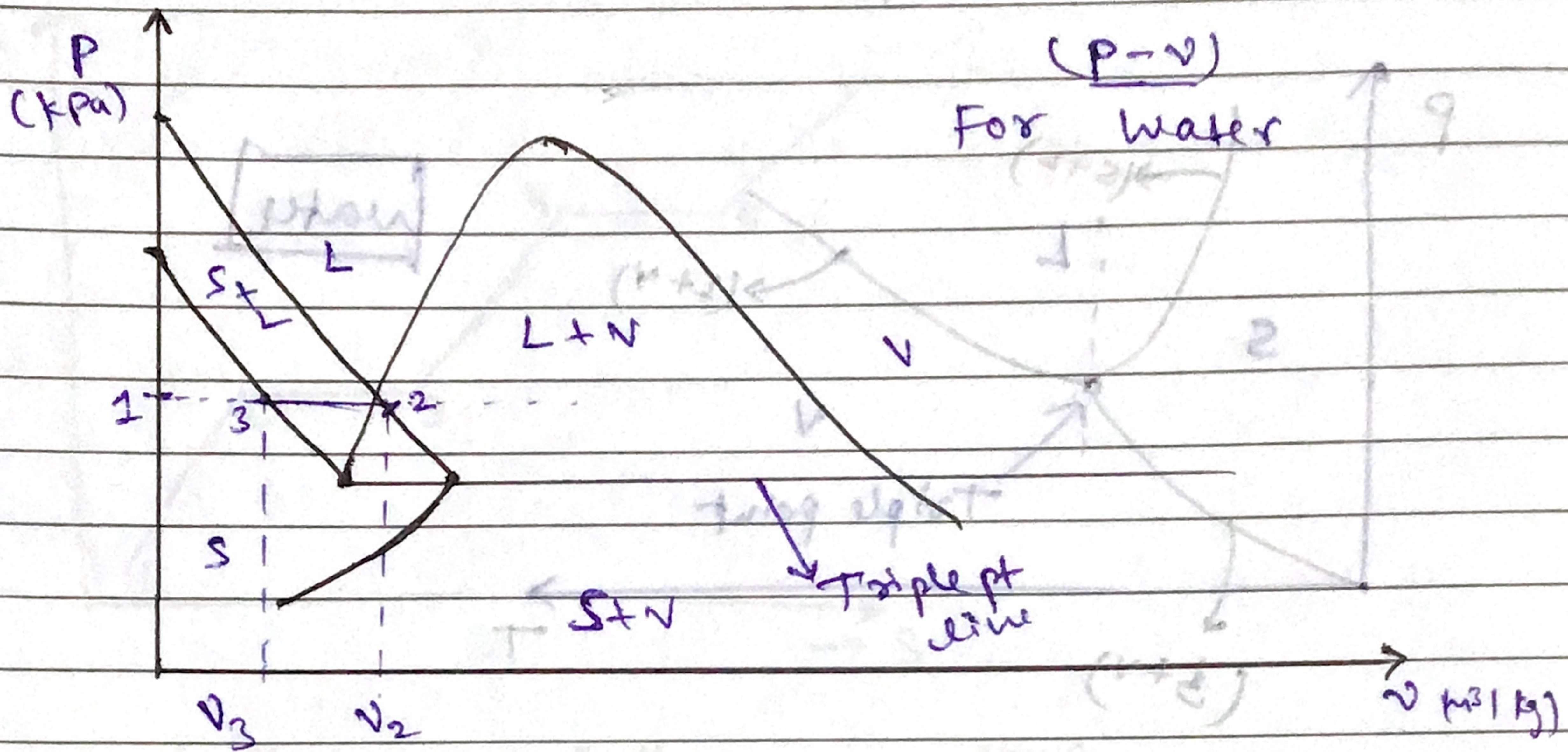
ΔT ← Change in temp

phase transformation



P-v graph





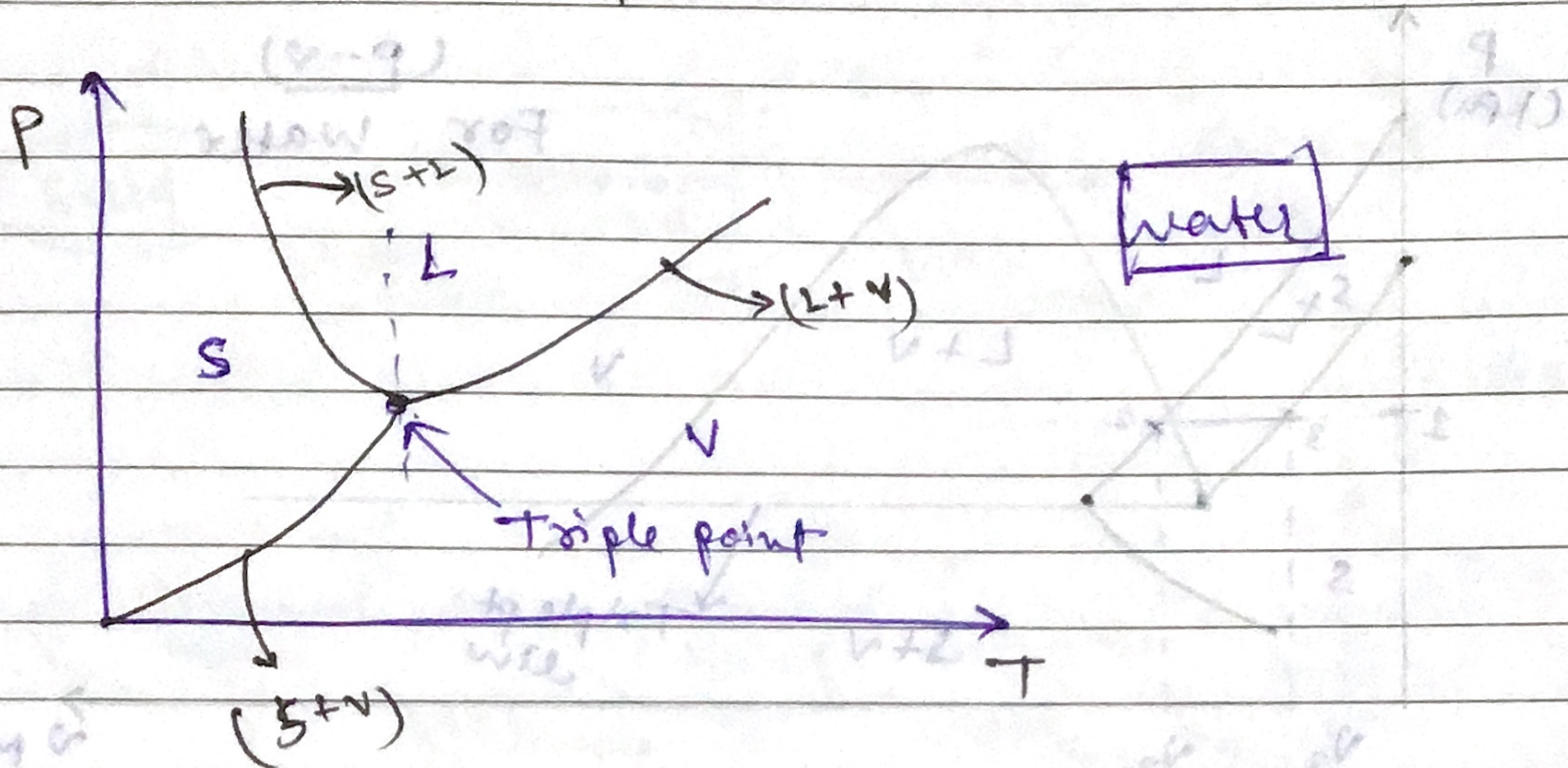
$v_3 < v_2$ ← this happens due to the hydrogen bonding. the sp. volume at 3 decreases

fig 1 → $\Delta v_L < \Delta v_V$ It is clear that the compression in liq is always less than the compression in vapor/gas

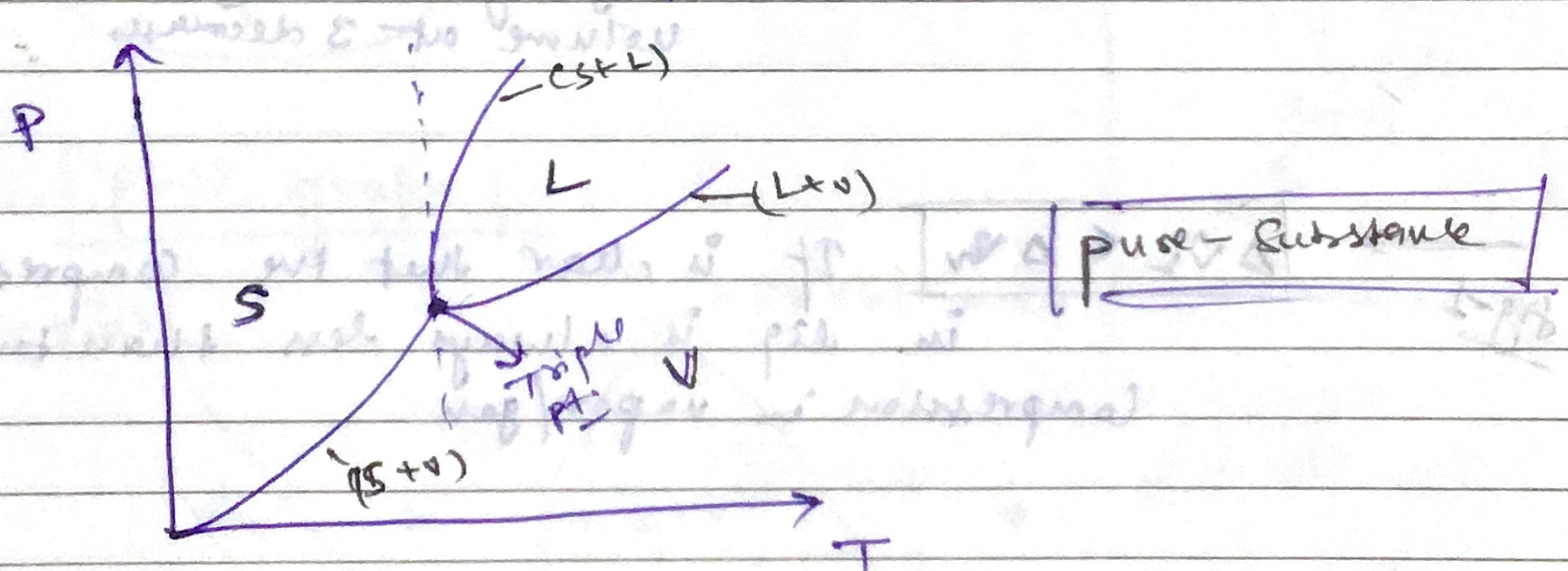
critical point → point at which direct transformation from $S \rightarrow V$ or $V \rightarrow S$ takes place with no intermediate fusion

Reason: As pressure increases, energy req. to change the phase decreases.

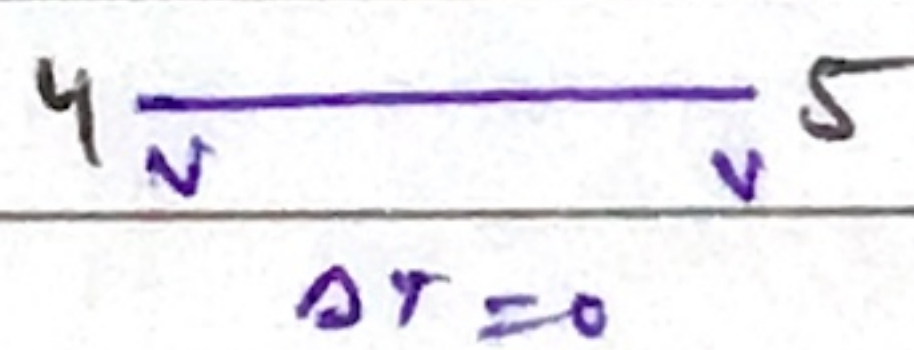
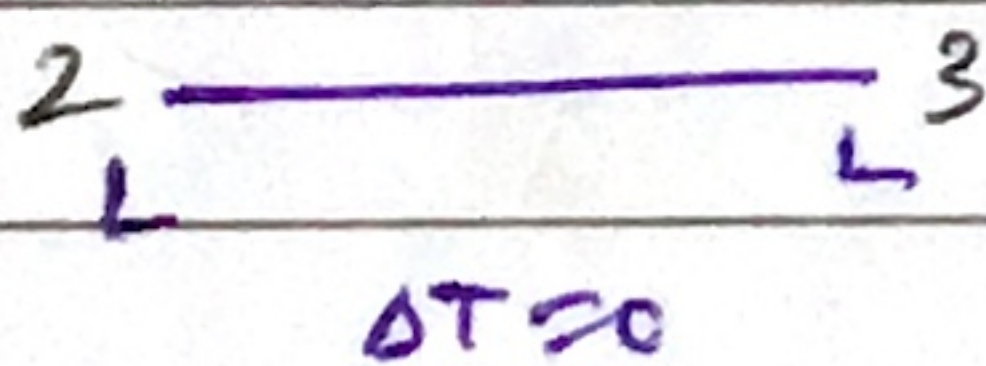
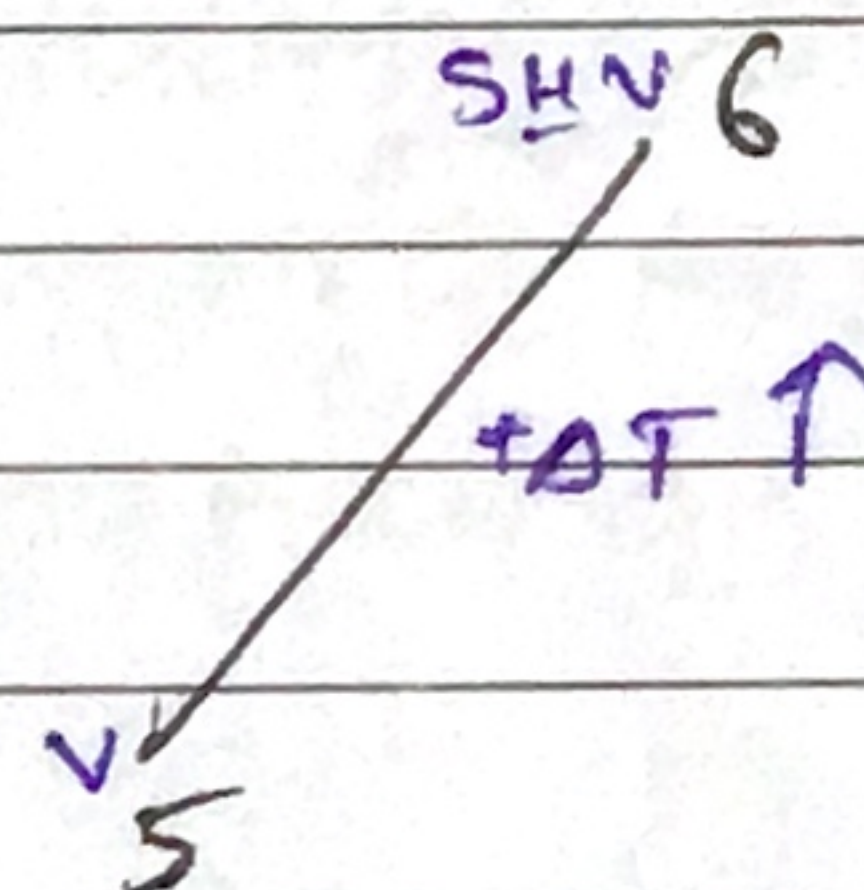
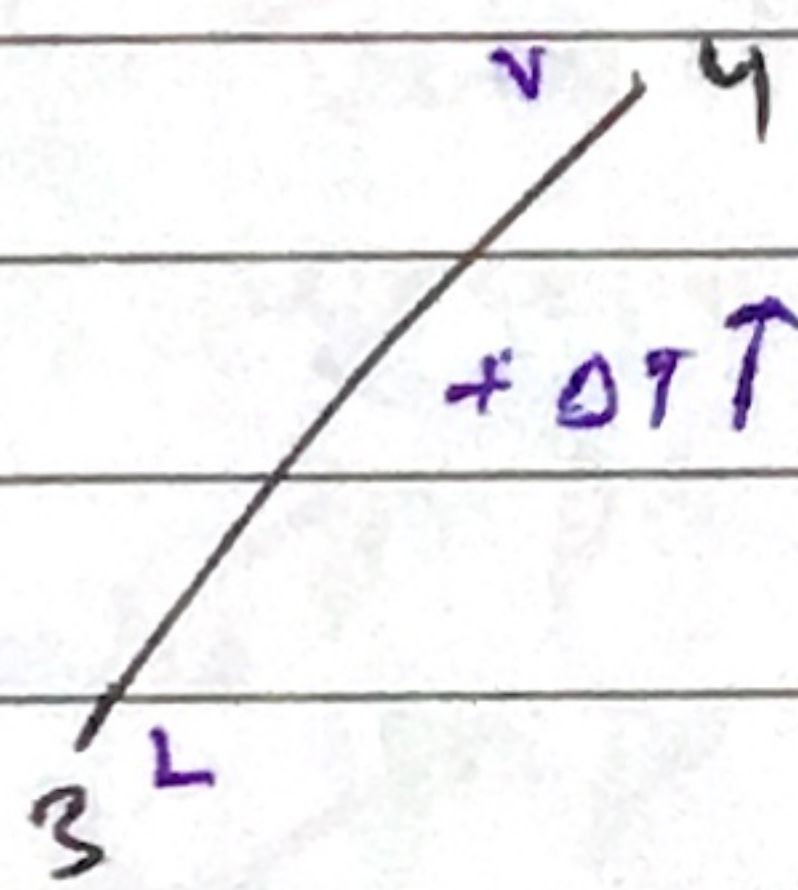
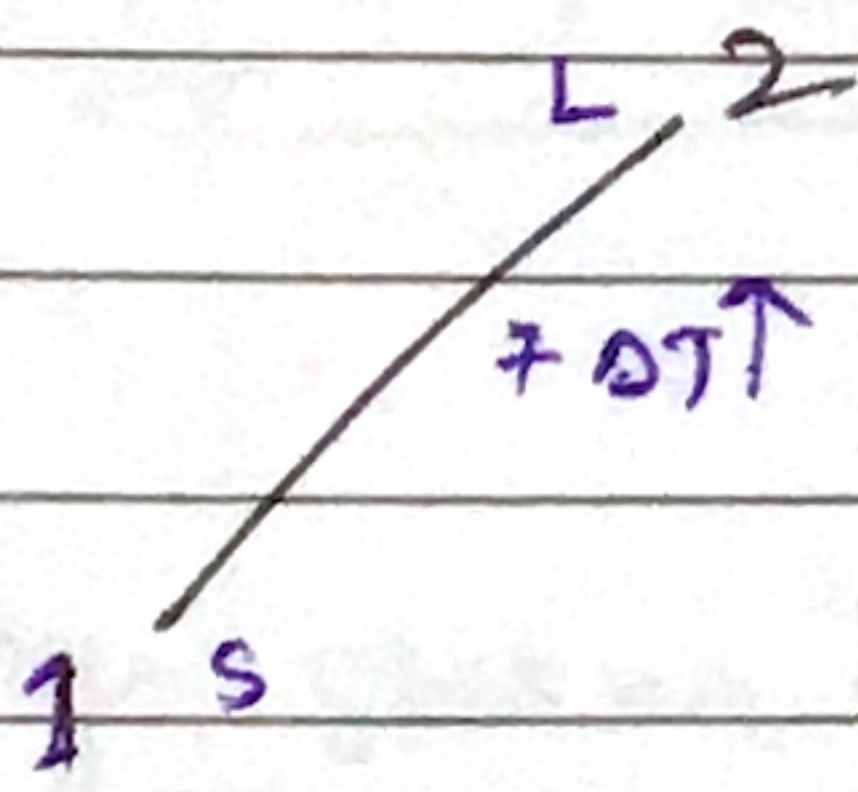
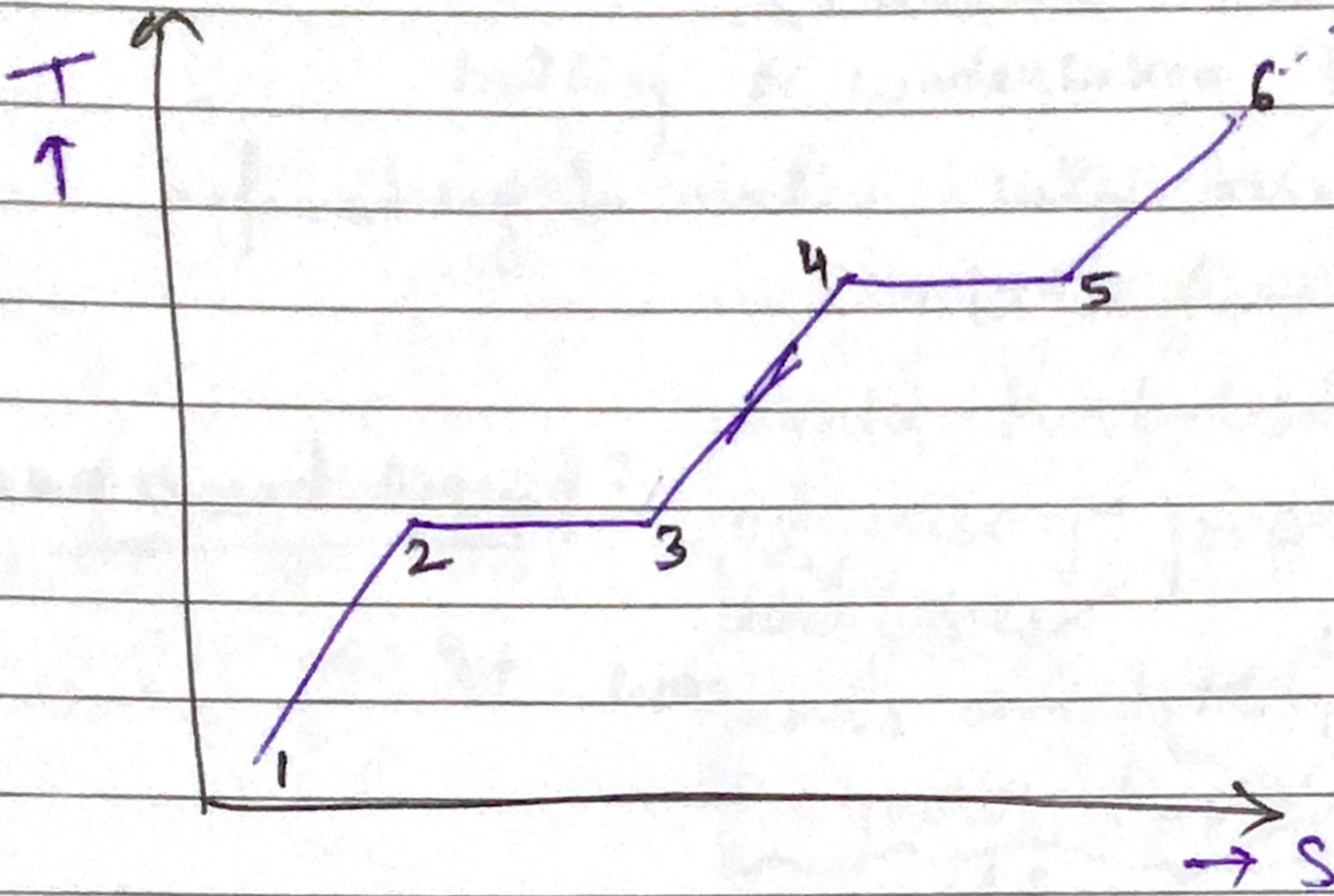
P-T graph for pure substance



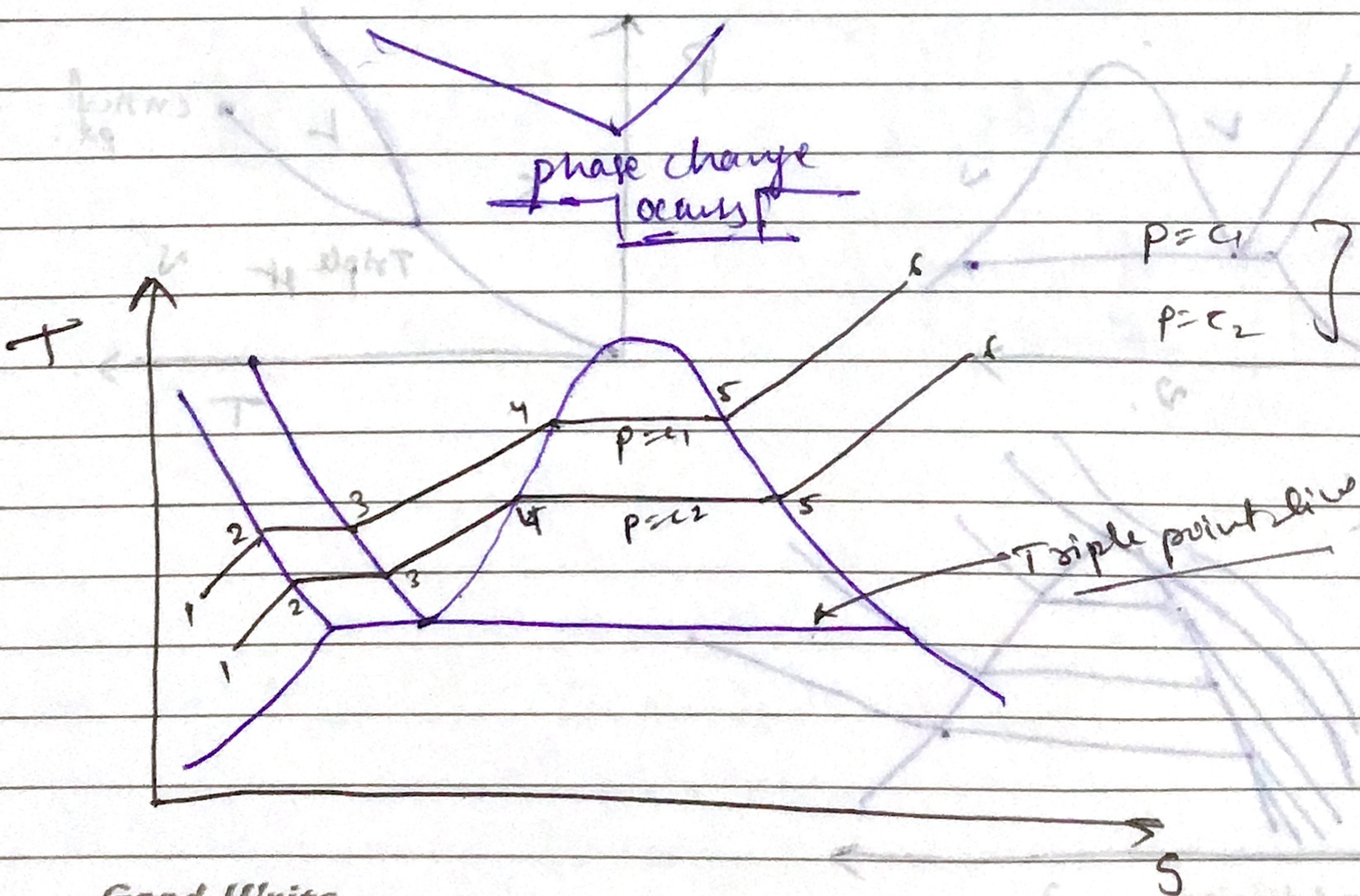
Triple point $\leftarrow$ point at which solid, liquid and vapour co-exist.



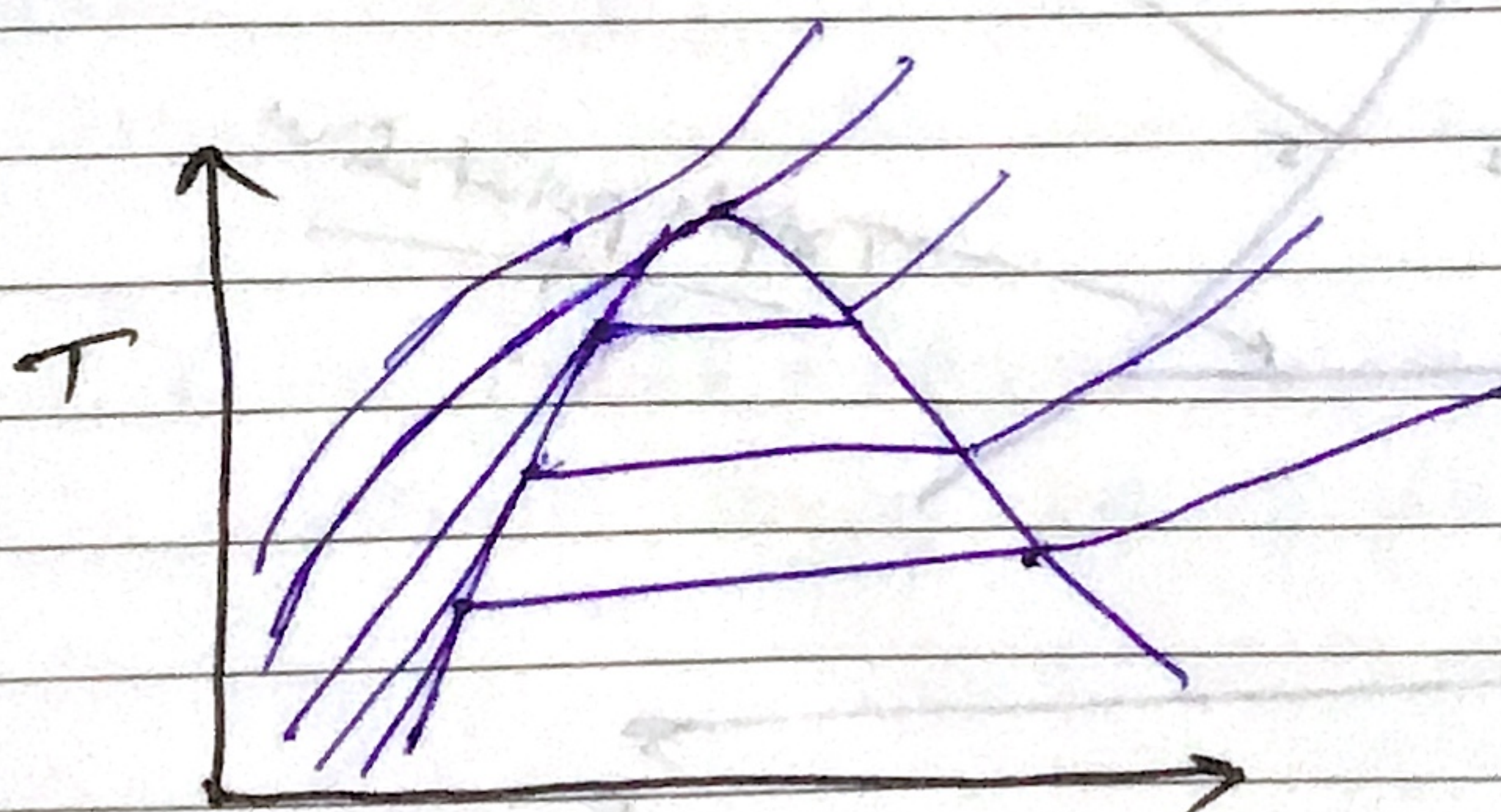
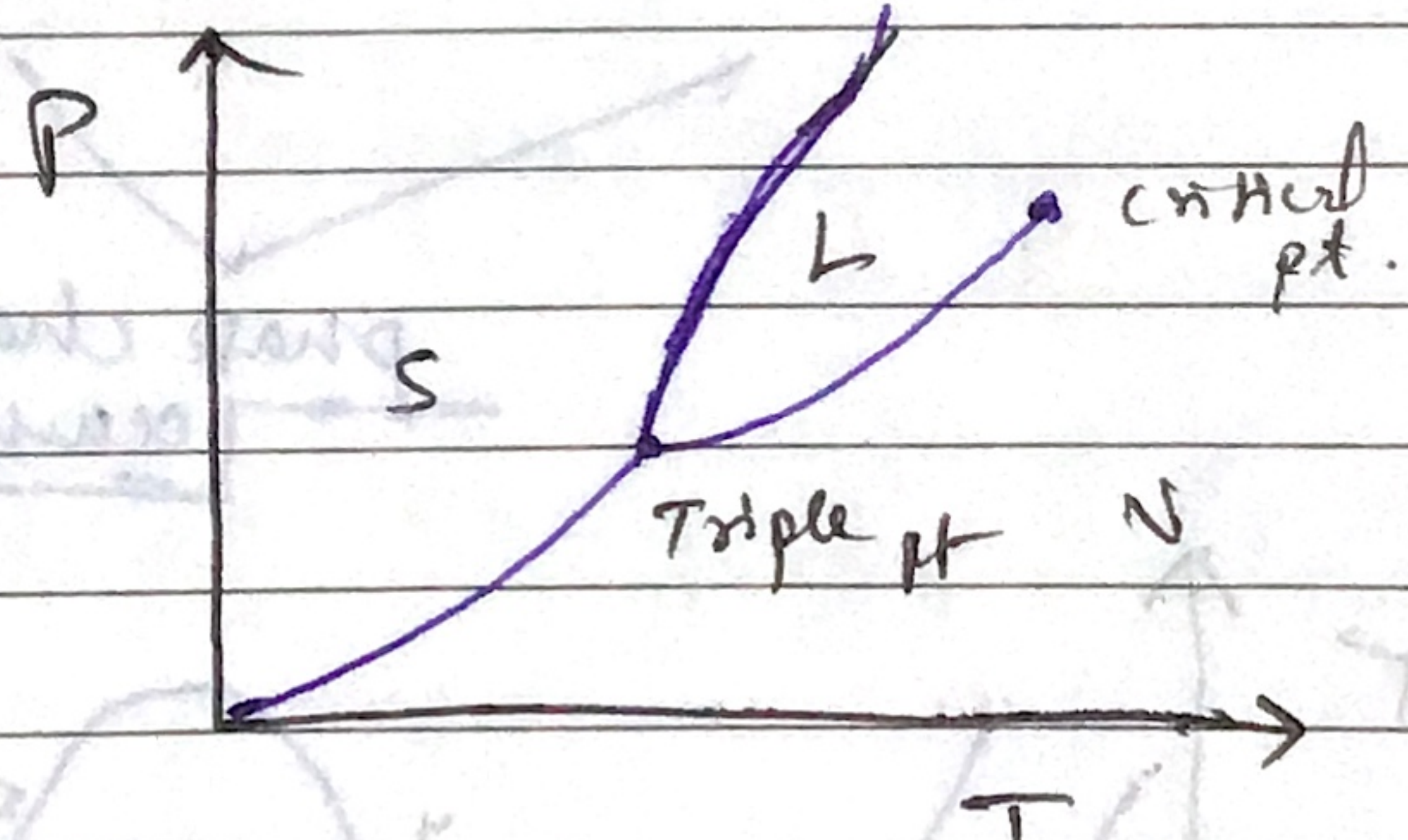
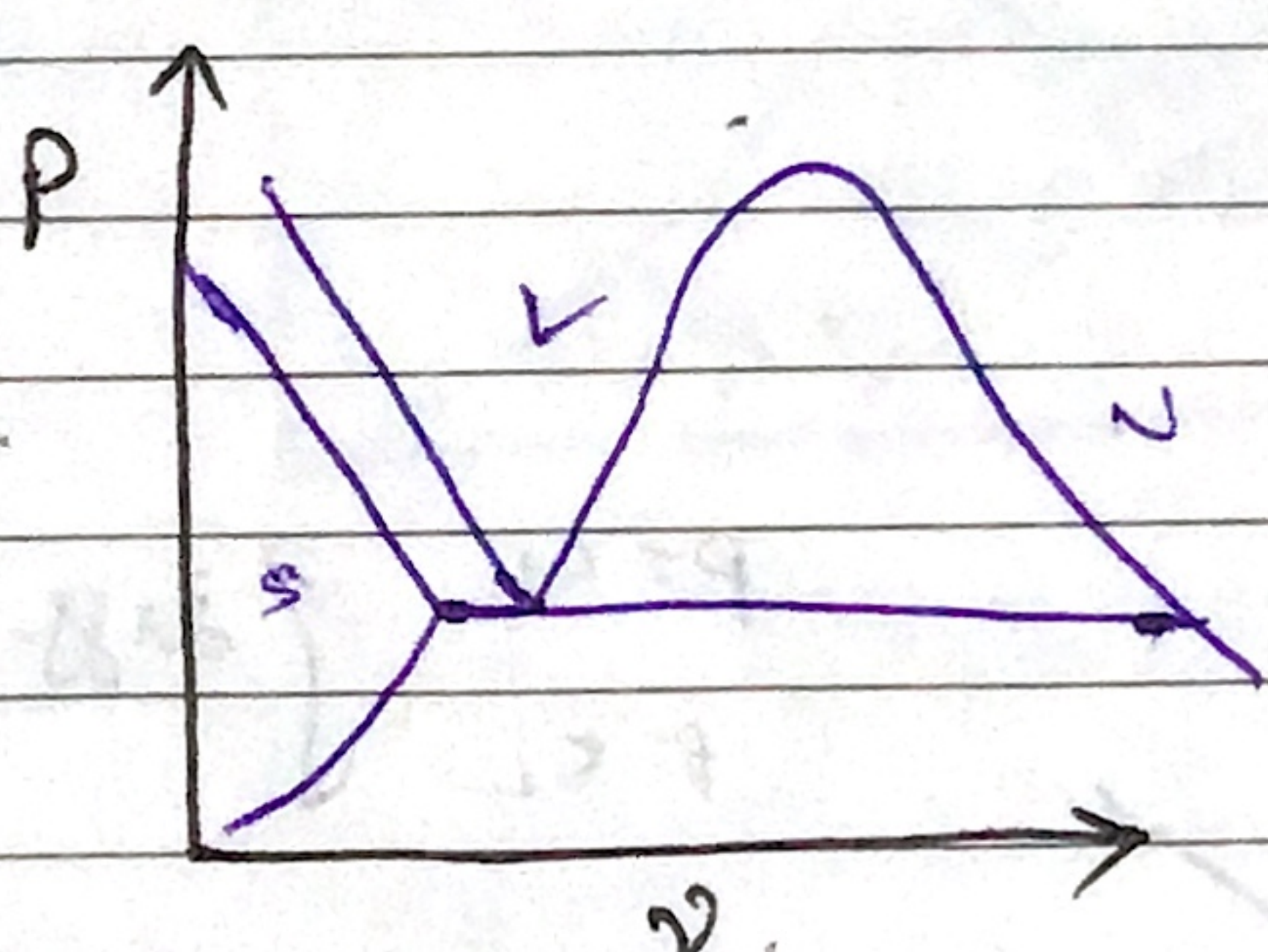
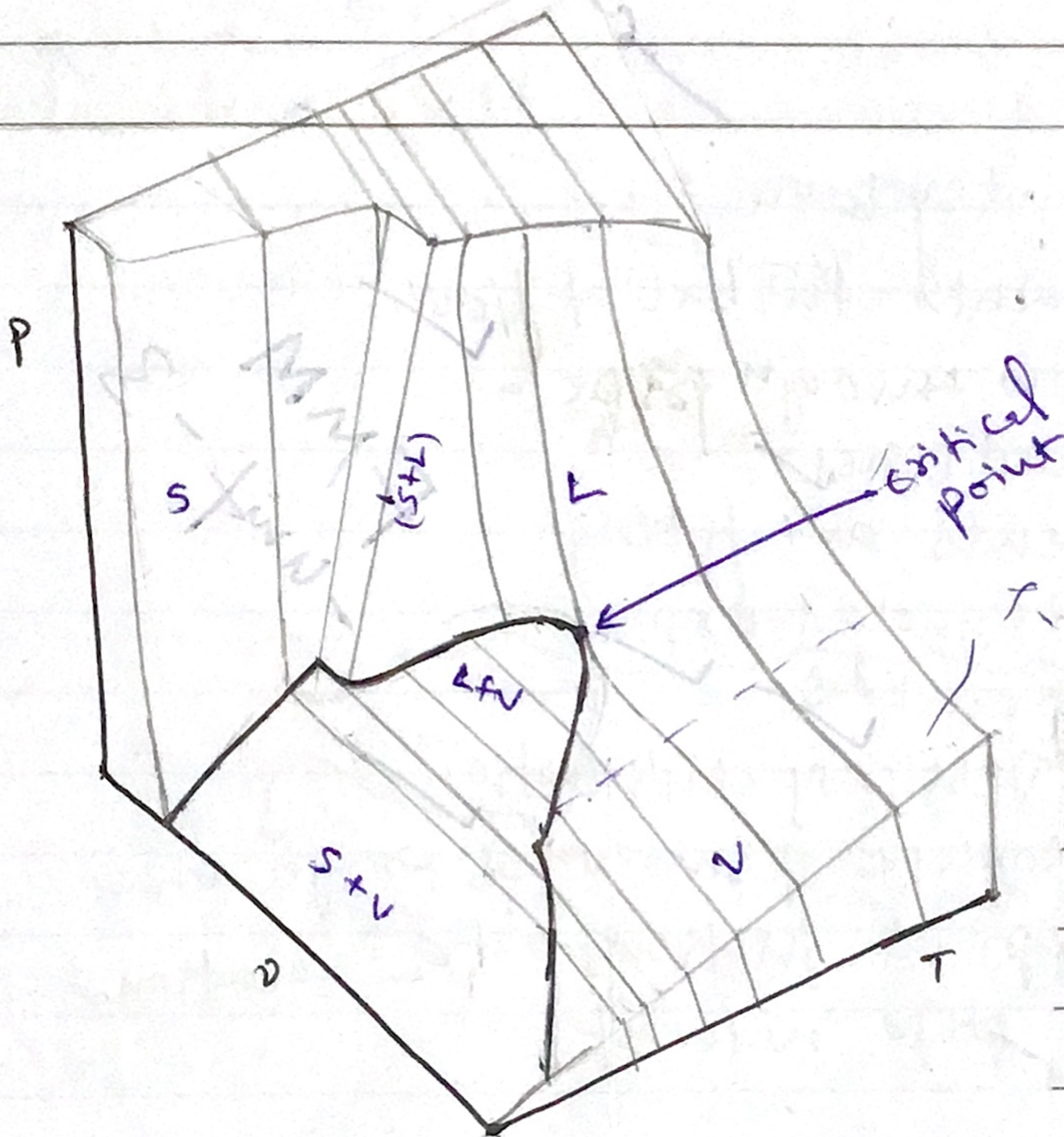
T-S diagram



phase change occurs



P-V-T diagram



Saturation state: Refers to the condition where a substance exist at its boiling or condensation temp.
any change in value will result in phase change either side.

Compressed liquid: Sub-cooled liq. a liquid below its saturation temp.
it would not go phase change.

Saturated steam: vapour existing at its saturation temp.

Dry steam: Saturated vapour with no liquid content.
 $x = 1$ — dryness fraction

$$u_g = u_b$$

$$2bT = h_b$$

$$T = \frac{h_b}{2b}$$

Mollier Diagram

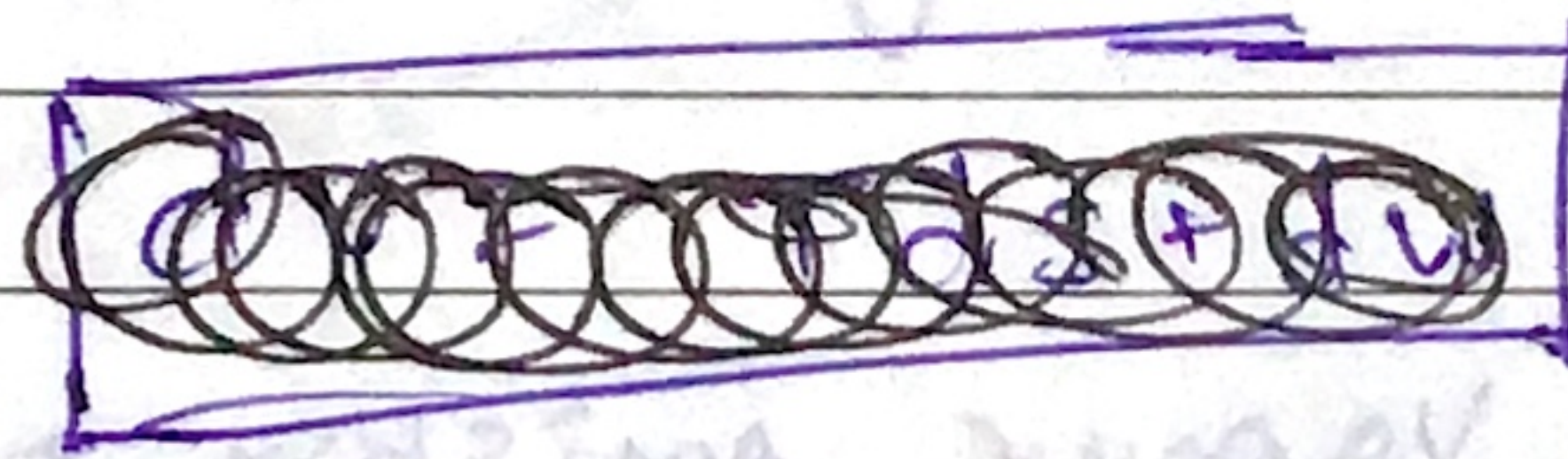
$$dQ = dU + dW$$

$$\Delta S = \frac{dQ}{T}$$

$$dQ = T ds$$

$$dU + dW = T ds$$

$dS = \Delta S$ just
considering
small change

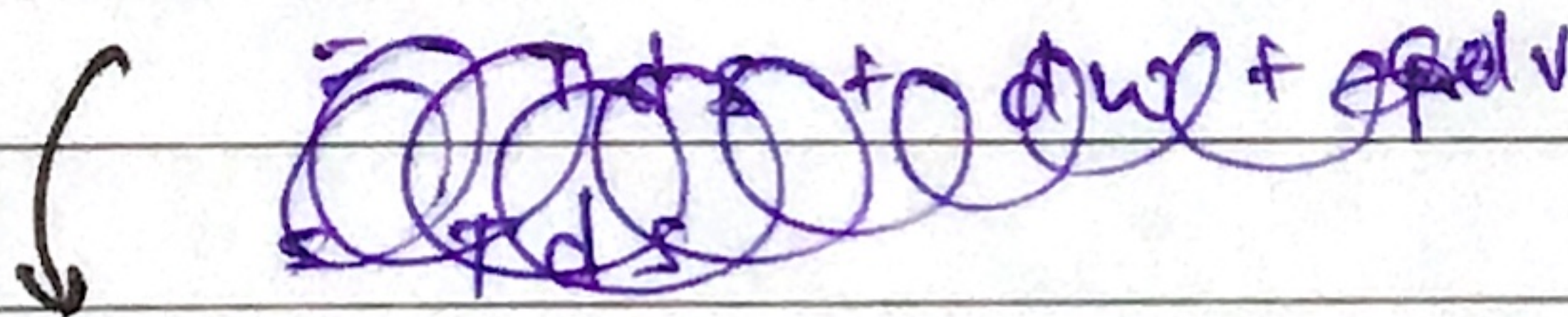


$$H = U + PV$$

$$dH = dU + p dv + v dp$$

$$dH = dU + p dv$$

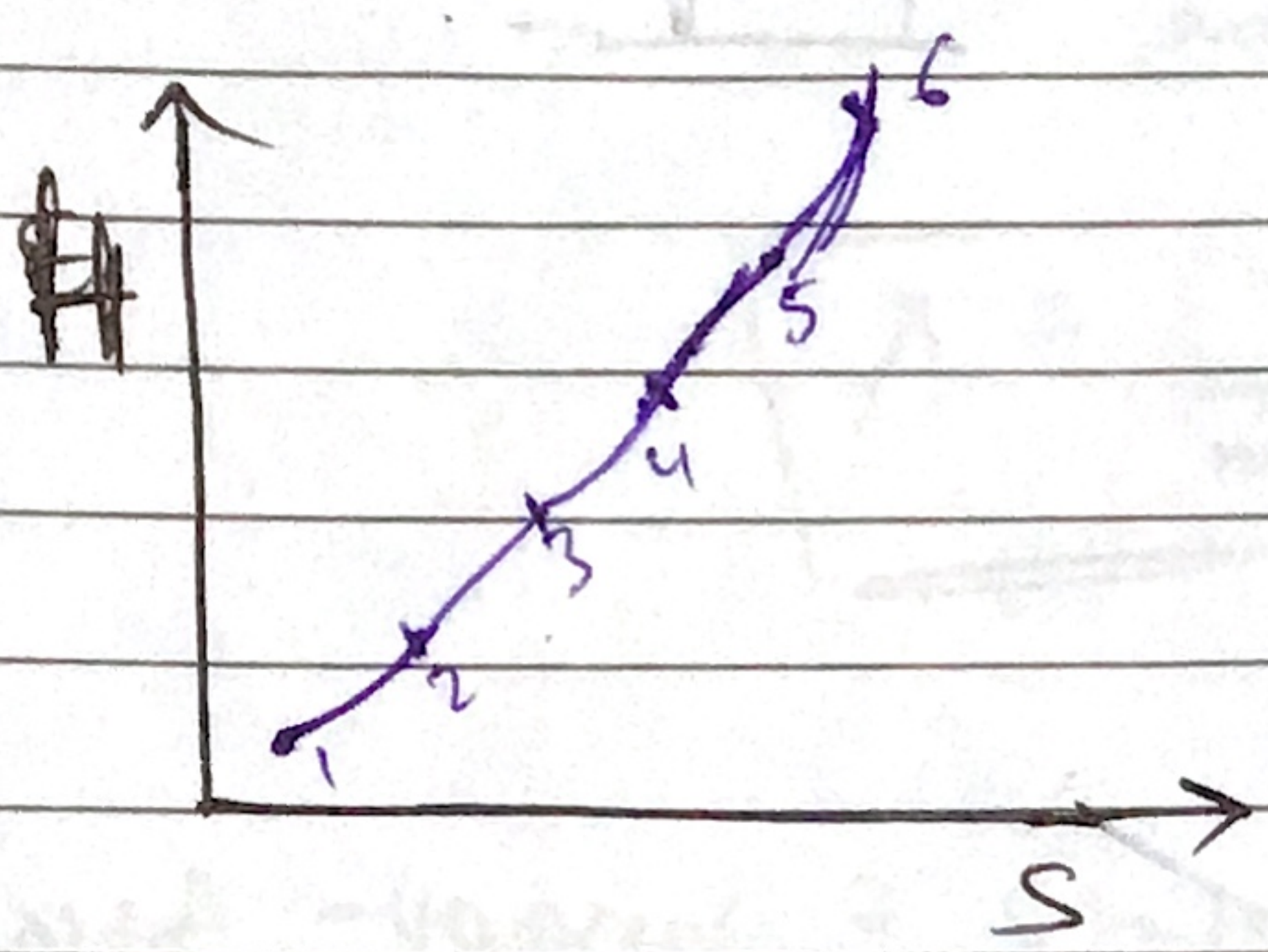
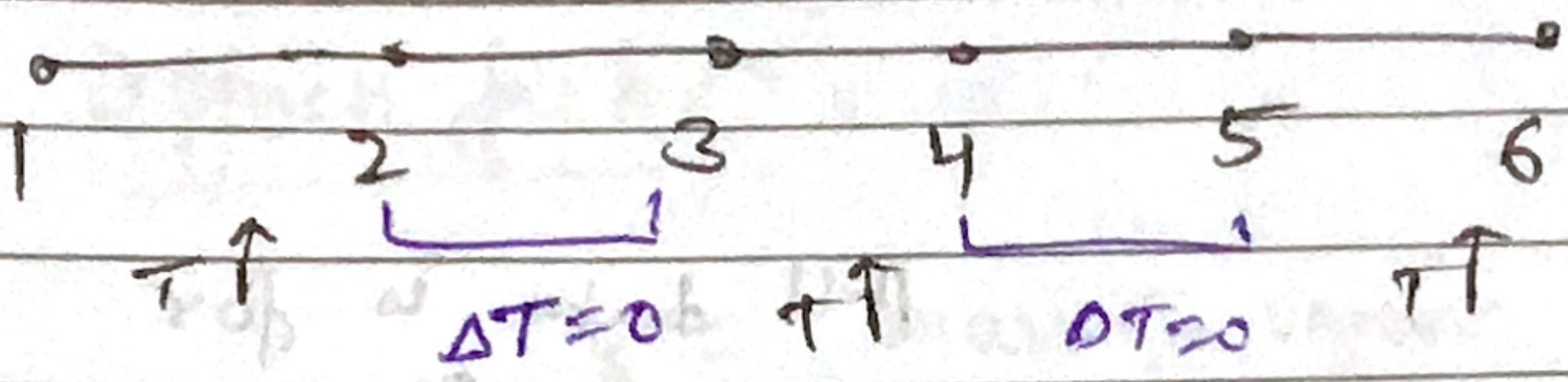
$dp = 0$
constant pressure



$$dH = T ds$$

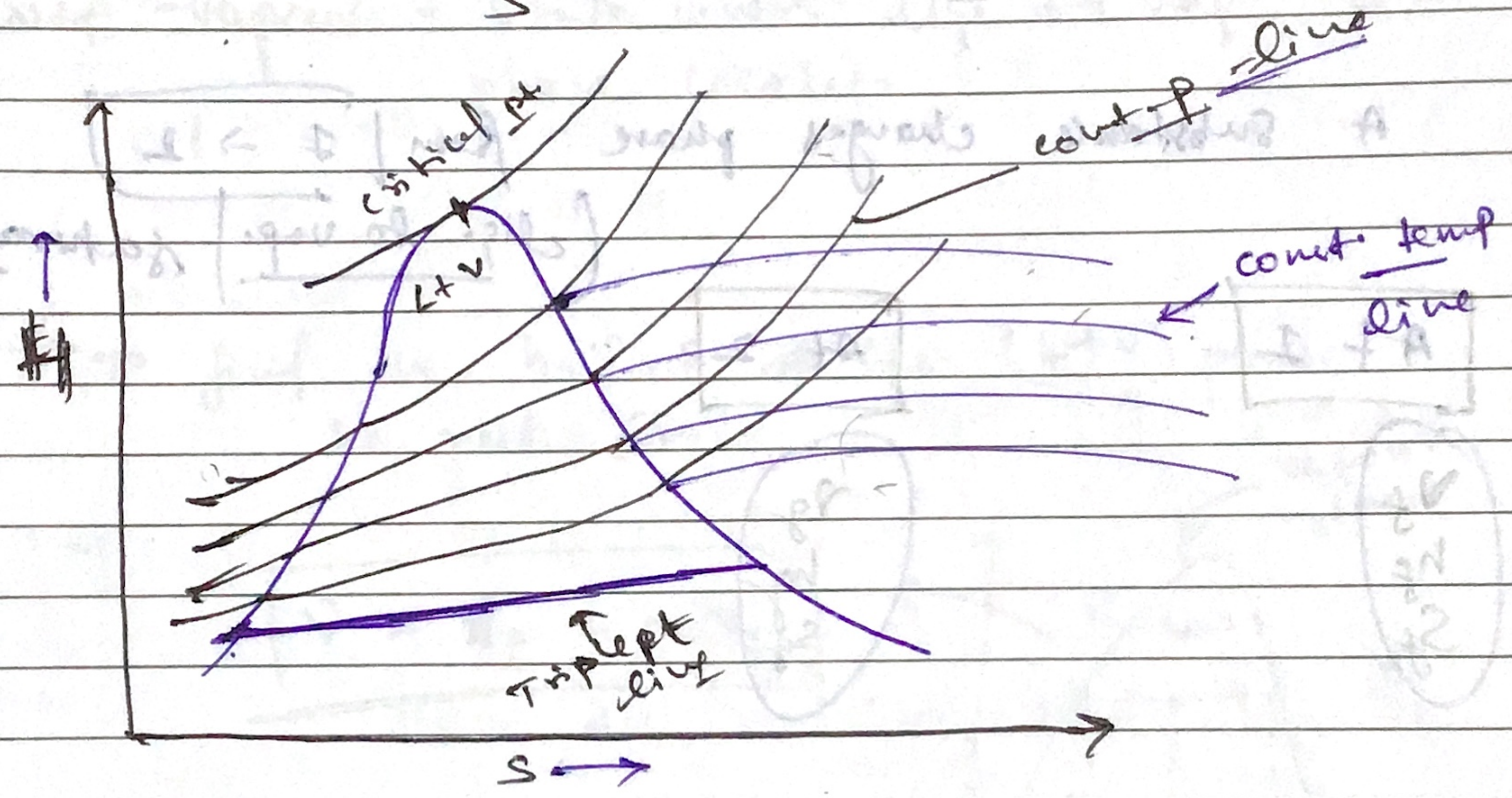
$$dW = p dv$$

$$\frac{dH}{ds} = T$$



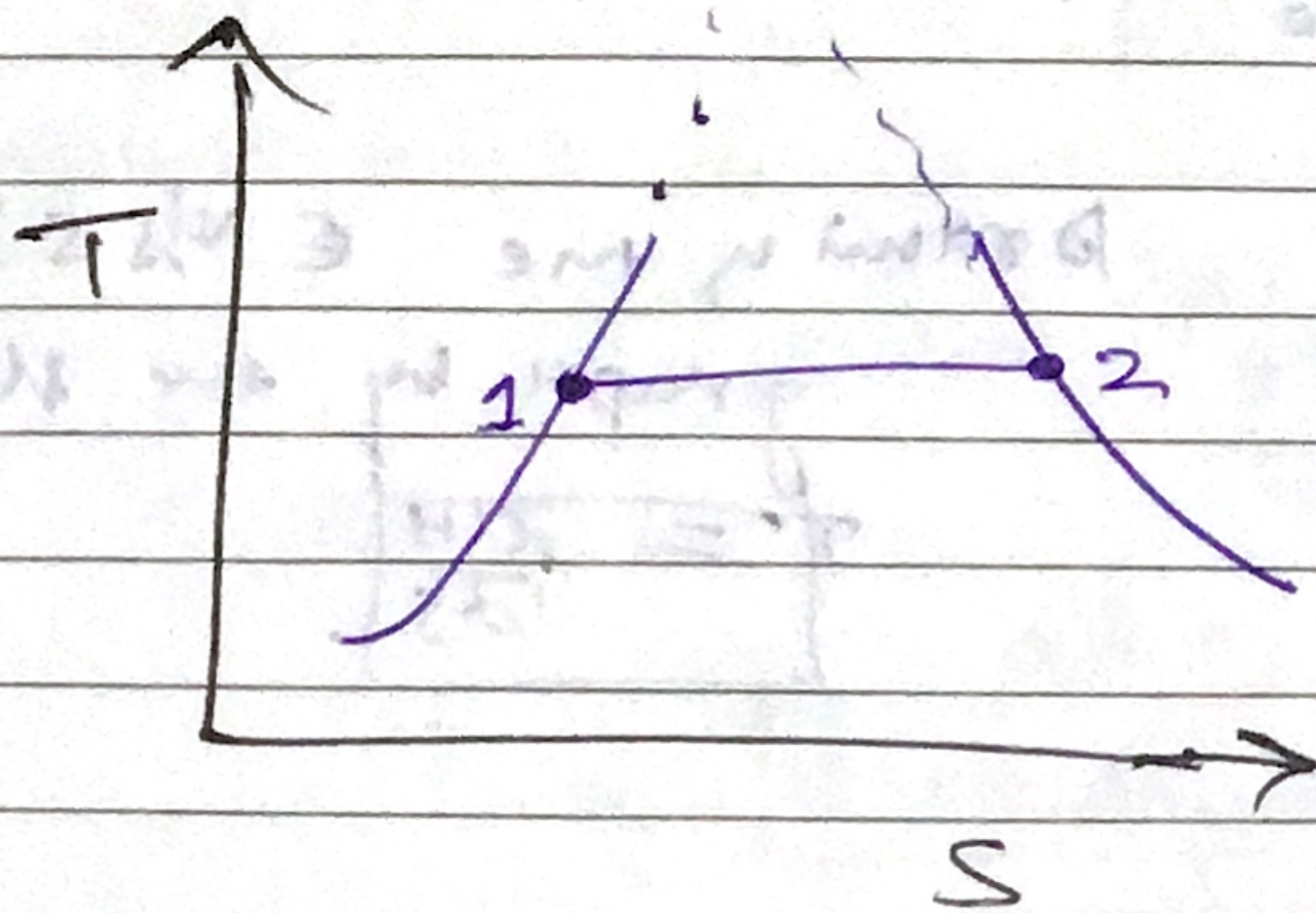
Drawing the E vs S graph by the slope.

$$T = \frac{dH}{dS}$$



Liquid + vapour

All data is for specific



A substance changes phase from 1 $\rightarrow$ 2
(liq. to vap.) saturated

At 1

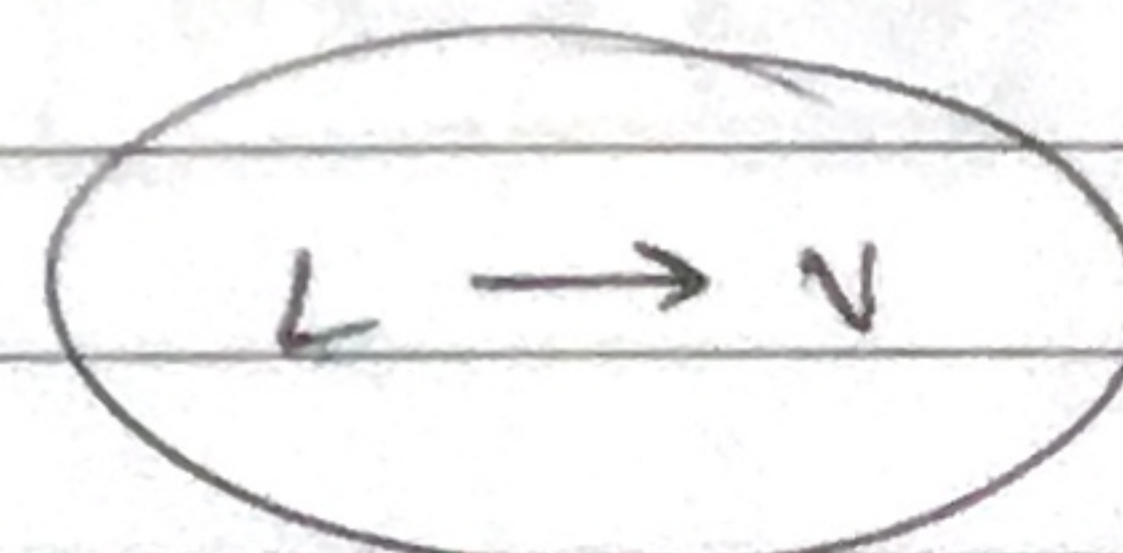
At 2

v_f
 h_f
 s_f

v_g
 h_g
 s_g

v_{fg} = Sp. vol of vapourisation

$$v_{fg} = v_g - v_f$$



Vapourisation process

$$h_{fg} = h_g - h_f$$

$$s_{fg} = s_g - s_f$$

Dryness factor

$$x = \frac{\text{mass of vapour}}{\text{mass of vapour} + \text{mass of liquid}}$$

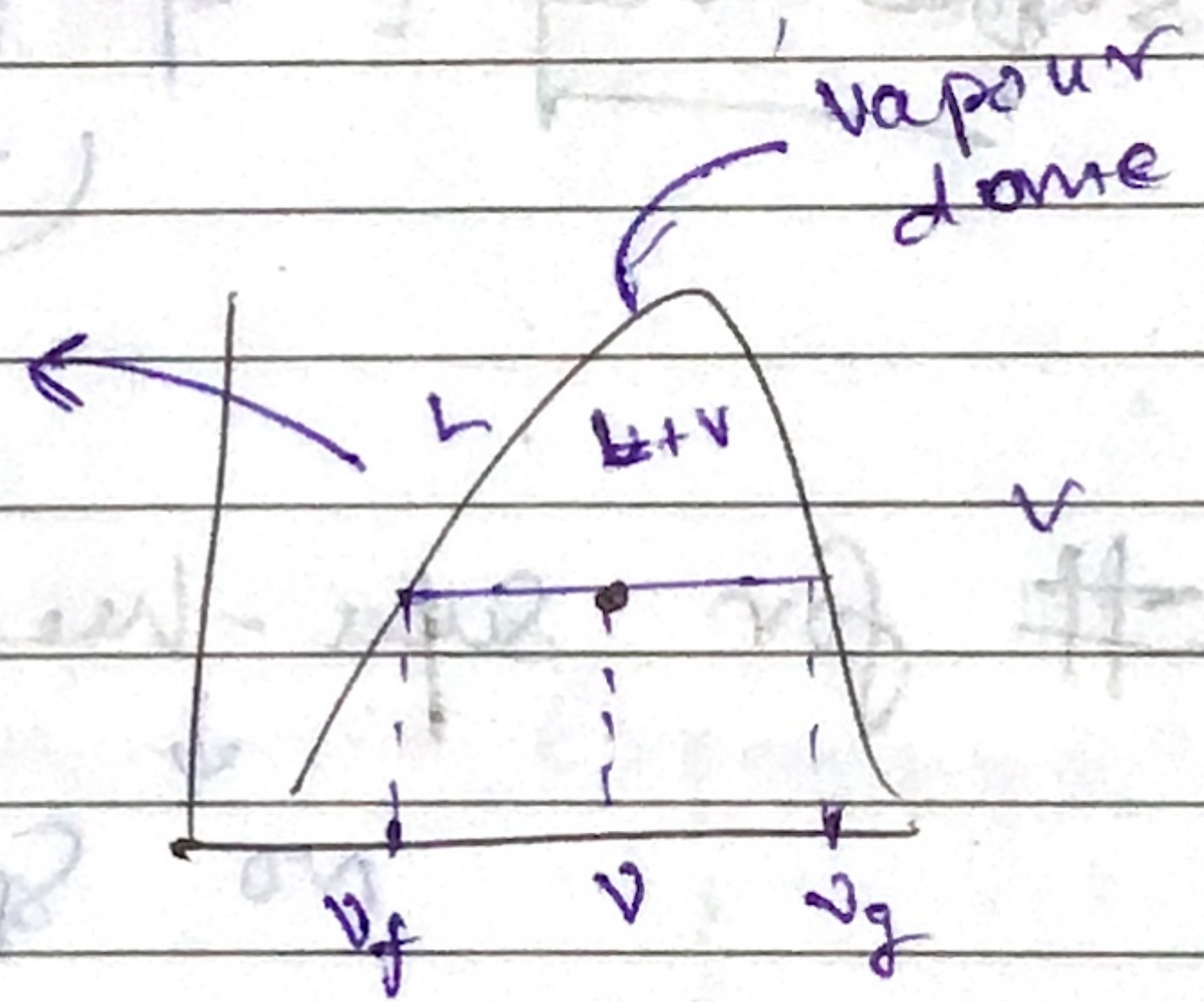
$$x = \frac{m_v}{m_v + m_l}$$

Wet-vapour : State where liq. and vap. both phase coexist.

→ to find the h, v, s at b/w (L+v) phase we need x .

eg.

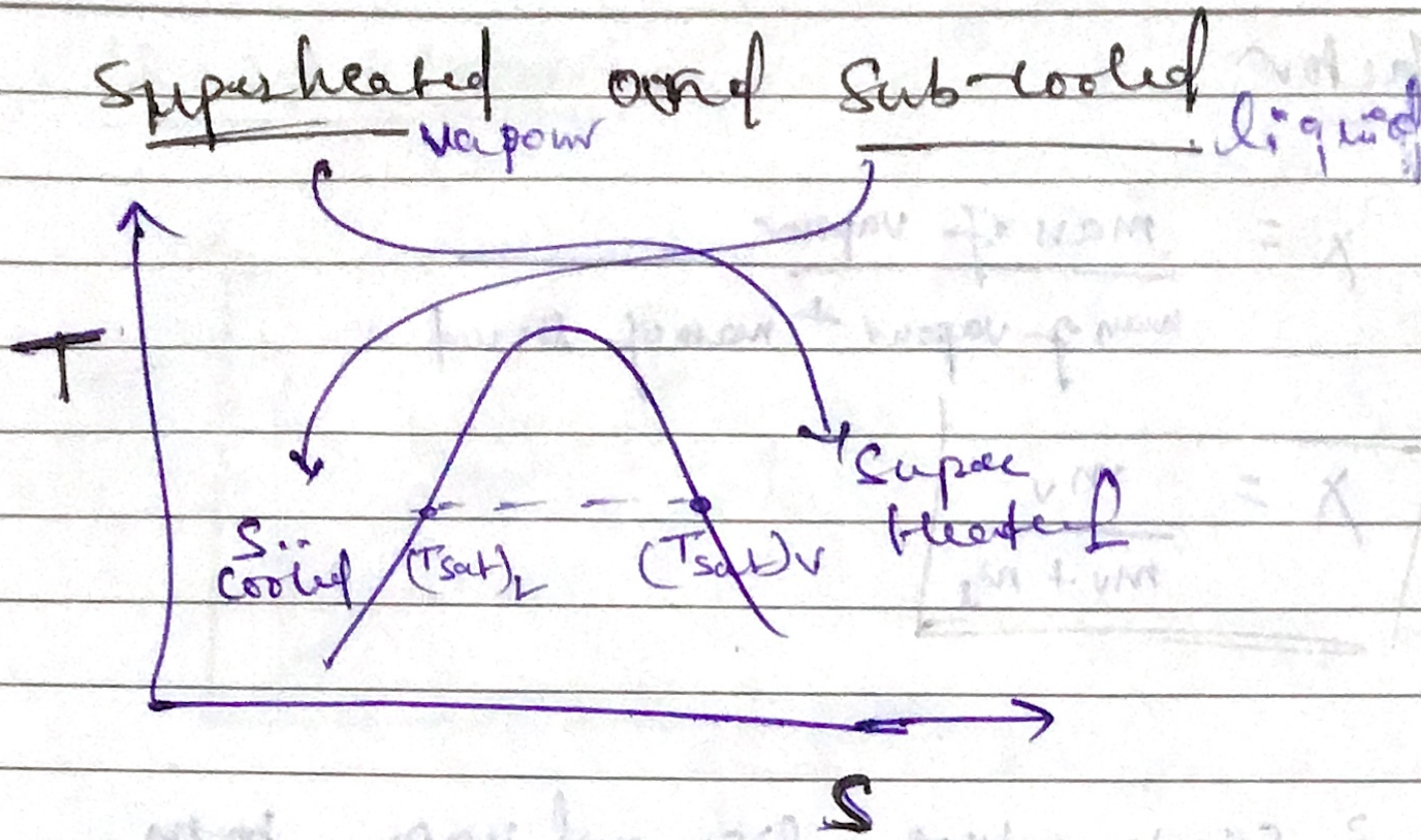
$$v = v_f + x v_{fg}$$



Table

T °C	Sat. pressure kPa	Specific volume	
		Sat. liquid v_f	Sat. vapour v_g
5	—	—	—
10	—	—	—
20	—	—	—
25	—	—	—

Table will be given
if values will be asked
by making "q"



In simple way $\rightarrow$ Superheated: no moisture.
 T is higher than $(T_{sat})_v$

Sub-cooled: T is lower than $(T_{sat})_L$

for Super-heated or Sub-cooled

no specific property is used
 normal prop. are used.

Gibbs phase Rule

$$F = C - P + 2$$

$F \leftarrow$ ° of freedom

$C \leftarrow$ no. of Component (constituent)

$P \leftarrow$ no. of phase in eqn.
 (S, L, V, G)